



ONTOLOGY REQUIREMENTS SPECIFICATION FOR AN EU DPP CORE ONTOLOGY PROPOSAL

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Abstract	This document outlines the requirements for an EU DPP core ontology proposal. It defines the DPP core ontology key terms and describes the methodology used to elicit the requirements for the ontology, leveraging concepts from various ontology engineering approaches, with Modular Ontology Modeling (MOMo) serving as the starting point. The document specifies the purpose of the ontology, its scope, and potential representation languages. It further details the identified ontology use cases, and functional and non-functional requirements. The functional requirements are articulated as competency questions. Key concepts, which may form the basis of ontology modules, are identified. The requirements and core terms of the EU DPP core ontology proposal are primarily derived from the ESPR and other regulatory documents. Distinct from MOMo, the requirements document also defines an initial set of core terms corresponding to these key concepts. In addition, the document addresses the requirements for evaluating and validating the DPP core ontology. This specification serves as an input for the design of an EU DPP core ontology proposed by the CIRPASS-2 Consortium.
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ABBREVIATIONS

DL	Description Logics
DPP	Digital Product Passport
ESPR	Eco-design for Sustainable Product Regulation
JSON-LD	JavaScript Object Notation for Linked Data
JSON	JavaScript Object Notation
LOT	Linked Open Terms
RDB	Relational Database Model
RDF	Resource Description Framework
R2RML	RDB to RDF Mapping Language
RML	RDF Mapping Language
SHACL	Shapes Constraint Language
SPARQL	SPARQL Protocol and RDF Query Language
UML	Unified Modelling Language
UNECE	United Nations Economic Commission for Europe
UN/CEFACT	United Nations Centre for Trade Facilitation and Electronic Business
UNTP	United Nations Transparency Protocol
OWL 2	OWL 2 Web Ontology Language
OWL DL	Web Ontology Language Description Logics
W3C	World Wide Web Consortium

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1 INTRODUCTION

This specification document serves as a guide for the development of an EU DPP cross-sectorial core ontology. We use the term “EU DPP core ontology” when referring to cross-sectorial ontology. We avoid using “DPP system ontology” since “DPP-system” is defined more broadly in ANNEX II of the CEN/CENELEC/ETSI standardization request¹: “DPP system means the set of IT standards and protocols required to ensure the full interoperability of product passports and compliance with the requirements referred to in point 2.1.”

In CIRPASS-2 User Stories 3.0 [van Nieuwenhuijze et al., 2024] document the term “DPP Core System” is defined as follows:

“DPP Core System means a minimal set of services (application services and other services) and the roles that deploy or perform these services, as required for compliance with the ESPR’s requirements for Digital Products Passports (e.g., Art. 9 and 10, ESPR²).”

The CEN/CENELEC/ETSI standardization request defines also the term DPP-Data as follows:

“DPP-data means the information included in a product passport and accessible to different users based on their own respective access rights.”

The information content to be included in product-specific DPPs will be defined in delegated acts of the ESPR or in other EU regulations.

In contrast to product-specific DPPs, the proposed EU DPP core ontology is a knowledge model for high-level descriptions of the EU DPP domain notions and relationships among them. It is intended to be used across all sectors to ensure semantic interoperability of the DPP system.

Consequently, the proposed EU DPP core ontology is related to DPP data and DPP system, but its goal is not to represent DPP system concepts and therefore focuses on DPP data concepts.

EU Vocabularies portal, which publishes many EU knowledge assets including common vocabularies, data models and ontologies, defines the term ontology as follows³.

“An ontology – in the context of computer and information sciences – can be defined as a formal specification designed to delimit and group instances/concepts (facts, events, entities, elements, etc.) based on their common class (types, properties, interrelationships, etc.), and thus formalizing a full or a subset of a domain.”

According to this definition, the CIRPASS-2 consortium defines this EU DPP Core ontology as follows.

The EU DPP core ontology is a shared formal specification of basic concepts and properties/relationships in the cross-sectorial DPP domain to facilitate knowledge sharing across sector-specific DPPs.

¹ [https://ec.europa.eu/transparency/documents-register/detail?ref=C\(2024\)5423&lang=en](https://ec.europa.eu/transparency/documents-register/detail?ref=C(2024)5423&lang=en)

² <https://eur-lex.europa.eu/eli/reg/2024/1781/oj>

³ <https://op.europa.eu/en/web/eu-vocabularies/ontologies>

In this document and in the CIRPASS-2 project, the following types of ontologies are distinguished (Fig. 1):

1. **The proposed EU DPP core ontology** introduces an extensible high level of abstraction of the EU DPP domain. It can be extended or specialised in sector-specific ontologies or modules while maintaining coherence and consistency. Should other cross-sectoral ontologies be proposed for the EU DPP, semantic mappings may be created from the proposed ontology to other ontologies, and vice versa.
2. **Sector-specific ontologies** extend the proposed EU DPP core ontology with sector-specific classes and relationships, and act as an interface between the proposed EU DPP core ontology and industry level ontologies.
3. **Industry level ontologies** include several low-level concepts that describe a certain aspect of the DPP data used in industry.

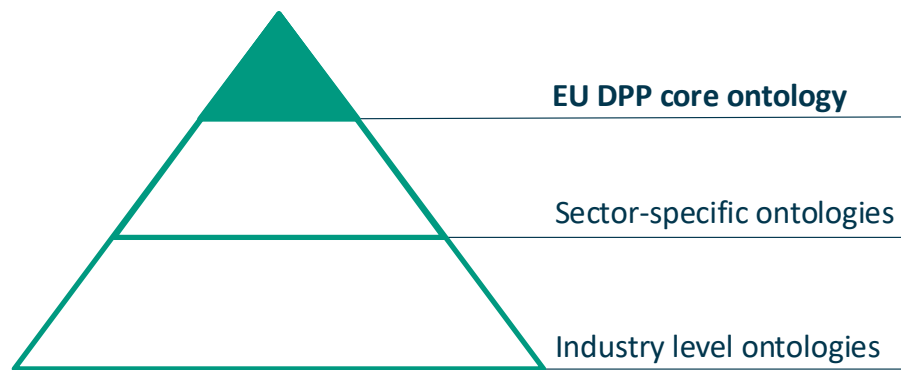


FIG. 1. TYPES OF ONTOLOGIES IDENTIFIED FOR THE CIRPASS-2 PROJECT.

This EU DPP core ontology is designed to enable semantic interoperability and knowledge sharing across different DPPs, fostering integration and reusability.

This document is consumed as an input for the design of the CIRPASS-2 EU DPP core ontology.

2 METHODOLOGY

No single-size-fits-all method exists for ontology development. Pattern-based methods [Blomqvist and Sandkuhl 2005] are widely used as well as modular ontology design [Kebede et al., 2024, Shimizu et al., 2023, Suárez-Figueroa et al., 2012] that enable creation of ontology networks. Recently, an industrial oriented ontology engineering framework Linked Open Terms (LOT) was proposed for developing ontologies used for linked data applications [Poveda-Villalón et al., 2022].

For the DPP core ontology development, we base our methodology on several existing ontology engineering methodologies. Our starting point is the usage of the modular ontology development methodologies [Kebede et al., 2024, Shimizu et al., 2023, Suárez-Figueroa et al., 2012] rather than building a monolithic ontology. Modular ontology engineering methods divide ontology into conceptually separated parts (modules) that are related to each other via relationships. In this specification, we essentially advantage from the Modular Ontology Modeling (MOMo) methodology, described by Shimizu et al. [Shimizu et al., 2023]. They define the concept of a module as:

“An (ontology) module is a part of an ontology which captures a key notion, and its key relations to other notions.”

They also emphasise that modules are guided, not only by the semantics of the domain but also by the development context and use case.

The benefits of using modular ontology development methodologies in development of an EU DPP core ontology includes a possibility to develop different modules separately by teams of competent area experts and ontologists. For example, Product, Units, LCA and Compliance are key notions that could be considered as the modules of an EU DPP core ontology that address some specific aspects of the DPP domain. In addition, ontology modularisation supports reusability of ontologies i.e., the modules of an EU DPP core ontology can be reused by other projects alike, as well as external ontologies can be reused in the form of a module. The MOMo methodology workflow [Shimizu et al., 2023] includes steps for ontology requirements elicitation as well as for ontology design and building/coding. The ontology requirements specification includes identifying ontology use cases, gathering competency questions (CQ) and identifying key notions. Ontology building steps start from identifying existing ontology reusable design patterns and creation of schema diagrams for modules. Ontology development documentation with embedded schema diagrams is an important part of the MOMo method. This is applied to modules as well as the whole ontology. Finally, based on the set of diagrams, an OWL file of ontology is created, published and used.

The current document presents only the ontology requirements specification and leaves the description of ontology building for specific specification documents (e.g., design specification).

According to the MOMo methodology workflow, ontology requirements specification document delivers the results of the following steps:

1. Identifying **ontology use cases**, resulting in a set of use case descriptions (brief textual summaries of use scenarios). Use cases are also called use scenarios in literature as in [Grüniger and Fox 1995].
2. Gathering of **CQs**, resulting in a list of CQs. Competency questions are widely used in ontology engineering, and they were first introduced by Grüniger and Fox [Grüniger and Fox 1995] for modelling enterprise ontologies.
3. Identifying **key notions** that are possible candidates for initial ontology modules and a set of core terms related to key notions.

In addition, principles of ontology engineering from [Gómez-Pérez et al., 2004] are applied to provide scope and goal of the ontology, a list of non-functional requirements and ontology representation languages.

The LOT ontology development methodology [Poveda-Villalón et al., 2022] gives also useful guidelines for ontology requirements gathering as well as ontology implementation. LOT is focused on merging ontology engineering and software development. In the LOT, special attention is paid to ontology publishing as this is very important for linked data consumption. Former ontology engineering methodologies did not consider it important.

Currently, the importance of ontology publishing is widely accepted by using Findable, Accessible, Interoperable and Reusable (**FAIR**) **ontology publishing and sharing principles on the web** [Le Franc et al., 2020, Poveda-Villalón et al., 2020].

CIRPASS-2 follows **FAIR principles** [Le Franc et al., 2020, Poveda-Villalón et al., 2020] in the proposed EU DPP core ontology development and adopts some ontology design methods proposed in the LOT methodology to better meet linked data requirements with respect to publication of ontologies.

To make ontologies, such as an EU DPP core ontology, findable, accessible, and (re)usable for responsible economic operators to implement or create a DPP, a vocabulary hub is required. According to the IDS-RAM

⁴⁴, the vocabulary hub is a service that stores, maintains, and publishes data models and ontologies while enabling collaborative management of these models.

Within CIRPASS-2, the **DPP Vocabulary Hub** has been developed at the domain <https://dpp.vocabulary-hub.eu> to publish the proposed EU DPP Core Ontology. This ensures the ontology is publicly available and reusable in combination with other models published in the DPP Vocabulary Hub, such as sectorial DPP ontologies. In essence, the vocabulary hub serves as a centralized hub for efficiently managing and publishing different knowledge assets, which are used to define the meaning of Digital Product Passports.

The proposed EU DPP core ontology will be developed through several iterations. After the first iteration, the ontology will be introduced to sectors and pilots (CIRPASS-2 WP3), which allows to identify and include cross-sectorial concepts present in at least two sectors to the proposed EU DPP Core ontology and having a potential value for new upcoming sectors to be included into the proposed EU DPP Core ontology.

3 ONTOLOGY USE CASE DESCRIPTIONS

In ontology specification methodologies, including MOMo [Shimizu et al, 2023], use case descriptions are brief textual summaries of ontology use scenarios that help to understand the problem to be addressed by the ontology, including potential future extensions.

The following ontology use case descriptions are identified for the proposed EU DPP core ontology in the CIRPASS-2 project:

- By providing a **shared understanding of key concepts, relationships and data structures**, that is compliant with relevant regulations, the proposed EU DPP core ontology together with sector-specific ontologies help to collect required DPP data from various data sources actors have, in a manner that ensures **compliance with relevant regulations** as well as interoperability. In this context, interoperability means that this information can be easily reused by other operators or consumed by applications, even if these operators or applications are in different sectors or go beyond minimum ESPR requirements.
- After the ontology is instantiated with data from the data sources, it must be able to **answer** different **queries from the intended end-users** of a DPP system. The intended end-users of the proposed EU DPP core ontology together with sector-specific ontologies are the value chain actors as listed in ESPR recital 32. Any end-user of the DPP will retrieve a subset of the DPP data artefact, based on their access rights. Example queries are addressed by the competency questions listed in the section describing Functional Requirements.
- The proposed EU DPP core ontology can be utilized in a web portal set up by the Commission to enable searching and comparing data within DPPs, while ensuring consistency with the respective access rights of the intended end-users.
- The proposed EU DPP core ontology **can be used to develop sector-specific ontologies**. Sector-specific ontologies either inherit from the core ontology or map to the core ontology, enabling semantic interoperability. Thereby, the proposed EU DPP core ontology represents a baseline that would be found in all sector-specific ontologies.

⁴⁴ https://docs.internationaldataspaces.org/ids-knowledgebase/ids-ram-4/layers-of-the-reference-architecture-model/3-layers-of-the-reference-architecture-model/3_5_0_system_layer/3_5_6_vocabulary_hub

- The established EU DPP Core Ontology can be **used by application developers responsible for building or integrating DPP applications**. Using a shared vocabulary helps to build interoperable, and regulation-compliant solutions.

4 PURPOSE AND SCOPE

The purpose of developing the proposed EU DPP core ontology is to provide a shared, regulation-compliant (i.e., ESPR) cross-sectorial formal specification for structuring and interpreting sector-specific knowledge derived from data and ensuring semantic interoperability of sector-specific DPPs.

The scope of the proposed EU DPP core ontology covers mandatory concepts of DPP required by ESPR. Voluntary concepts related to DPP may be considered in the future as extensions of the core ontology.

The proposed EU DPP core ontology can serve as a semantic anchor (a way to dynamically link, e.g., via URI, a term/property to other vocabularies) for sector-specific ontologies and other DPP related ontologies by third parties or jurisdictions.

Integration capacities (or options for linking) with external DPP ontologies that are currently under development by other organisations, research communities or projects will be considered and analysed during the design phase of the ontology. These are related to the UNECE UNTP activities in DPP ontology development, and other international DPP-related standardisation activities (ISO/TC 154, ISO 20435, IEEE P3828, etc.), as well as integration possibilities with dictionary systems. Integration requirement is associated to the extendibility of the product passports (Standardisation request 5423 (12) “*The extendibility of the product passports should cover at least: (1) additional type of data/properties/attributes and (2) additional records.*”).

5 ONTOLOGY REPRESENTATION LANGUAGES

At the conceptual level, the proposed EU DPP core ontology is represented using formal semantic languages like Web Ontology Language (OWL) or Resource Description Framework (RDF), which support machine-readable representations, reasoning, and interoperability.

Specifically, the proposed EU DPP core ontology is created in the OWL 2 Web Ontology Language⁵ (OWL 2). OWL 2 is the W3C standard that enables rich ontology representation and ontology consistency checks using Description Logic (DL) reasoners (e.g., OWL-DL reasoner Pellet⁶ or HermiT⁷). In addition, using OWL 2 for ontology representation makes it possible to use the semantic query language for RDF, SPARQL⁸, as well as Shapes Constraint Language (SHACL⁹) that is a language for validating RDF graphs against a set of constraints. Both SPARQL and SHACL, are W3C standards.

OWL 2 has several variants (e.g., OWL Full, OWL Lite and OWL DL), from what we use OWL 2 DL, which is a subset of OWL 2 providing favourable computational properties¹⁰.

⁵ <https://www.w3.org/TR/owl2-overview/>

⁶ <https://github.com/stardog-union/pellet>

⁷ <http://hermit-reasoner.com>

⁸ <https://www.w3.org/TR/sparql11-query/>

⁹ <https://www.w3.org/TR/shacl/>

¹⁰ <https://www.w3.org/TR/owl2-syntax/>

However, it is possible to specify the usages of the OWL 2 DL profiles called OWL 2 EL, OWL 2 QL, and OWL 2 RL¹¹ that are subsets of the OWL 2 DL to meet requirements of specific applications. The main goal of these profiles is to support efficient reasoning by providing restrictions of the usage of the OWL 2 DL language constructs. The EL acronym reflects that the profile is based on Existential Language (EL) family of description logics that provide only Existential quantification. The QL acronym represents that query answering in this profile can be implemented by rewriting queries into a standard relational Query Language (QL). The RL acronym expresses that reasoning in this profile can be implemented using standard Rule Language (RL). The required level of expressiveness of the proposed EU DPP core ontology representation is determined during the design phase of the ontology development.

Using OWL 2 DL for ontologies allows for multiple different serialisations leading to a faster and easier adoption. OWL 2 DL ontologies can be converted to RDF and Turtle formats (W3C standards), as well as to JSON-LD¹² (<https://json-ld.org/>), which is a Linked Data format based on JSON.

At the technical level, the semantic standards used for representation of the core ontology and its extensions can be transformed into representations like JSON-LD or Unified Modelling Language (UML) diagrams. These representations are practical for implementation in APIs or software specifications. For example, a JSON-LD representation derived from the core ontology ensures semantic integrity while enabling seamless API integration.

Several mapping languages from existing standards to semantic standards exist. For example, the interoperability with JSON is addressed by using open-source languages like RDF Mapping Language (RML)¹³. RML is defined as a superset of the W3C-recommended mapping language, RDB to RDF Mapping Language (R2RML¹⁴), that maps data in relational databases to RDF.

According to FAIR principles, **human oriented ontology documentation** is very important. For that purpose, the ontology representation should include ontology metadata and natural language annotations. Documentation that is external to ontology may include human readable ontology documentation in HTML, conceptualisation diagrams, mapping, use case examples, and etc. There is no standard convention for ontology diagrams and visualisation. Therefore, developers have adopted several approaches, such as UML-alike diagrams or custom diagrams. In [Poveda-Villalón et al., 2020] some guidelines for generating ontology diagrams based on the UML are provided.

6 ONTOLOGY REQUIREMENTS

6.1 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe expected features of a system that by nature are not behavioural, address performance, or set constraints. The following non-functional requirements have been identified:

1. The names of ontology classes, relationships and properties as well as their descriptions are presented in English.
2. Any ontology or vocabulary for DPP is created in the OWL 2 Web Ontology Language, convertible to other formats, e.g., JSON-LD.

¹¹ <https://www.w3.org/TR/owl2-profiles/>

¹² <https://www.w3.org/TR/json-ld/>

¹³ <https://rml.io/specs/rml/>

¹⁴ <https://www.w3.org/TR/r2rml/>

3. The ontology must be based on ESPR, Standardisation request 5423 (optional), and consider, if possible, relevant European regulations and standards in existence or under development.
4. The expandability of the product passports should cover at least: (1) additional type of data/properties/attributes, and (2) additional records (Standardisation request 5423 (12)) (optional). The extendibility requirement should be considered in the proposed EU DPP core ontology development.
5. Optionally, the product passport should also allow the inclusion of information concerning integrated product components (Standardisation request 5423 (3)).
6. The proposed EU DPP core ontology requirements consider discovering cross-sectorial concepts from sectorial level information defined for batteries, electronics and textiles sectors in the CIRPASS deliverable D2.1 [Wagner et al., 2023], “Data Attribute Longlist¹⁵” by the Battery Pass project, and the CIRPASS-2 D3.2 deliverable [O’Regan and Saklampanakis 2025].
7. To enable semantic interoperability, the ontology concepts and the related EU DPP core vocabulary terms could be equipped with a ‘semantic anchor’. A semantic anchor represents a semantic relationship to the nearest term in some other common vocabulary enabling to establish a connection with other ontologies and vocabularies. For example, for the proposed EU DPP core ontology terms, the Schema.org¹⁶ ontology as well as the UNTP DPP web vocabulary could be used as a semantic anchor.
8. External ontologies used within the proposed EU DPP core ontology should go through rigorous evaluation and validation process before the usage. This concerns among other evaluation criteria also checking of broken links to ontology modules or external ontologies used.
9. For the needs of the CIRPASS-2 project, the DPP Vocabulary Hub is developed at the domain <https://dpp.vocabulary-hub.eu> to publish the proposed EU DPP Core Ontology. This hub is hosted and owned by one of the CIRPASS-2 consortium partners (TNO). This ensures the ontology created by the CIRPASS-2 project is publicly available and reusable in combination with other CIRPASS-2 project models published in the DPP Vocabulary Hub, such as sectorial DPP ontologies. In essence, the vocabulary hub serves as a centralized hub for efficiently managing and publishing different knowledge assets. The vocabulary hub needs to support OWL 2, RDF, and JSON-LD at minimum.
10. The proposed EU DPP core ontology covers at least mandatory concepts of a DPP required by ESPR, with a possibility to extend these with additional concepts.
11. An abstract class will be defined and used whenever the value, its type and structure are a matter of sectorial information, may differ cross-sectors, or is not yet defined by a delegated act.
12. From organisational point of view, it is important that Public Authorities are delegated to be responsible for the proposed EU DPP core ontology maintenance to guarantee that ontology will be available, up-to-date, consistent, etc.

6.2 FUNCTIONAL REQUIREMENTS

¹⁵ [https://thebatteryass.eu/assets/images/content-guidance/pdf/2023 Battery Passport Data Attributes.xlsx](https://thebatteryass.eu/assets/images/content-guidance/pdf/2023_Battery_Passport_Data_Attributes.xlsx)

¹⁶ <https://schema.org>

The functional requirements (behaviour of the ontology), following the MOMo methodology, are described through Competency Questions (CQ). CQs make it possible to validate whether an ontology supports the end-user needs.

6.2.1 COMPETENCY QUESTIONS

Competency questions (CQ) are considered as end-user requirements in the form of examples of natural language queries for which the proposed EU DPP core ontology together with a sector-specific ontology should provide answers after the ontology is instantiated with data.

The proposed EU DPP core ontology together with sector-specific, industry level and external ontologies will create an ontology network. CQ response accuracy can be evaluated only after instantiation of the proposed EU DPP core ontology related ontology network (e.g., hierarchy) with data property values (instances, data points) obtained from data sources (e.g., provided by the CIRPASS-2 pilots).

Note that these CQs are relevant only if required in the **corresponding delegated act**. The list provided below is not exhaustive as it is a proposal.

- CQ1:** What are the values of all or selected mandatory attributes of a DPP required by ESPR and delegated acts at the granularity level specified in the delegated act (model, batch or item level) or at the model level?
- CQ2:** What are the values of all or a selection of performance parameters (e.g., repairability score, durability score, carbon footprint, environmental footprint) of the product? (ESPR Art 7 (2b), (i), see ESPR Annex I for a list of possible product parameters).
- CQ3:** What type of digital instructions (e.g., install, use, maintain, repair, end-of-life) are available for the product, and how can they be accessed? (ESPR Art 7 (2b), (ii), (iii) and (iv), Art 7 (5d) and Art 7 (5e), Annex III (f))
- CQ4:** What are the names and numeric codes of substances of concern present in the product (ESPR Art 7 (5a), (i) and (ii), (iii), (iv)), and what is the location of substances of concern within the product? (ESPR Art 7 (5b))
- CQ5:** What is the concentration, maximum concentration or concentration range of the substances of concern, at the level of product/product's components? (ESPR Art 7 (5c))
- CQ6:** What are the components the product is made of? (ESPR Art 7 (5c))
- CQ7:** What are the spare parts available for the product? (ESPR Art 7 (5c))
- CQ8:** What is the Global Trade Identification Number (GTIN) of the product? (ESPR Annex III (c))
- CQ9:** What are the relevant commodity codes of the product? (ESPR Annex III (d))
- CQ10:** What are the compliance documents required under ESPR and delegated act(s) for a product?
- CQ11:** What is the manufacturer's unique operator identifier of a product? (ESPR Annex III (g))
- CQ12:** What are the unique operator identifiers of the economic operators connected to DPP? (ESPR Annex III (h))
- CQ13:** What are the unique facility identifiers of locations or buildings involved in a product's value chain or used by actors involved in a product's value chain? (ESPR Annex III (i), Art 2 (33))
- CQ14:** What is the name, registered trade name or registered trademark, postal address and electronic means of communication of value chain actors (including economic operator and importer)? (ESPR Art 29 (3), Annex III (j))

- CQ15:** What is the reference of the digital service provider hosting the back-up copy of the digital product passport? (ESPR Annex III (l))
- CQ16:** What is the date the product was placed on the EU market and what is the duration of validity period of the DPP? (ESPR Art 9 (2i))
- CQ17:** What are the access rights (introduce, modify or update) assigned to data item in the product DPP for value chain actors on the model, batch or item level? (ESPR Art 9 (f, g), Art 11 (f))
- CQ18:** What are the life-cycle events that have occurred associated with the product?
- CQ19:** What is the previous DPP of a remanufactured product?
- CQ20:** What are the weight and volume of the product and its packaging, and the product-to-packaging ratio? (ESPR Annex I (j))
- CQ21:** What is the available life-cycle assessment-related information for the product?
- CQ22:** What is the name, registered trade name, postal address and electronic means of communication of the DPP service provider?

The ontology functional requirements will be completed after verification performed during ontology maintenance.

7 EVALUATION AND VALIDATION REQUIREMENTS FOR THE PROPOSED EU DPP CORE ONTOLOGY

Ontology evaluation is extremely important in the context of networked ontologies that the proposed EU DPP core ontology together with sector-specific, industry level and external ontologies will create.

Different contexts and points of views to domains of interest may cause existence of incompatible axioms in an ontology network. Each ontology that is included to an ontology network should be priorly evaluated and validated.

The evaluation of ontologies refers to the correct building of the content of the ontology, i.e., ensuring that its definitions (a definition is written in natural language and in a formal language) correctly implement ontology requirements and competency questions, or perform correctly in the real world. Ontology validation refers to whether the meaning of the definitions really models the real world for which the ontology was created [Gómez-Pérez 1996, 2004].

Several methods that provide automated or semi-automated evaluation and validation of ontologies are reviewed and compared in [Raad and Cruz 2015]. In addition, the Good Ontology Design Guideline (GoodOD) is proposed to improve quality of ontology development [Boeker et al., 2012]. The ROME methodology, which identifies requirements that an ontology must satisfy, and maps these requirements to evaluation measures, is presented in [Yu et al., 2009].

We drive evaluation and validation of the proposed EU DPP core ontology from the requirements. This process includes evaluation of functional and non-functional requirements presented in this document.

7.1 EVALUATION OF THE PROPOSED EU DPP CORE ONTOLOGY CONTENT

Criteria-based evaluation approaches use criteria to evaluate several characteristics of ontologies by manual inspection of an ontology by domain experts and ontology engineers. Manual work is needed as not all

criteria may be easily quantifiable. If ontology is represented in the formal language like OWL 2, then some tools can be used to evaluate ontology.

Evaluation criteria for the proposed EU DPP core ontology content include consistency, completeness, and conciseness [Gómez-Pérez 1996, 2004] as follows:

1. **Consistency** means that the ontology does not include or allow for any contradictions. As we represent the proposed EU DPP core ontology in OWL 2, then the consistency of the ontology is evaluated using DL reasoners like Pellet.
2. **Completeness** measures if the domain of interest is appropriately covered in an ontology. CQs given in the requirements specification help to evaluate completeness. However, completeness cannot be proven. CQ completeness is fully applicable if the whole set of CQs is implemented. In our case, the CQ set is open for additions. The MOMo method also supports incremental addition of CQs. CQs can be evaluated by DPP domain experts and ontology engineers. In the case of OWL 2 ontologies, CQs can be represented by the corresponding SPARQL queries, and their response accuracy can be evaluated. CQ response accuracy can be evaluated only after instantiation of the proposed EU DPP core ontology related ontology network (e.g., hierarchy) with data property values (instances, data points) obtained from data sources (e.g., provided by pilots).
3. **Conciseness** criteria state if the ontology includes irrelevant elements with regards to the domain to be covered. Conciseness is hard to evaluate but domain experts/reviewers using their background knowledge could find irrelevant elements that do not belong to the domain under consideration. For example, redundancies of “subclass-of” and “instance-of” relationships and identical formal definitions of classes or instances can be detected.

7.2 ASSESSMENT OF EXPANDABILITY OF THE PROPOSED EU DPP CORE ONTOLOGY

Expandability (monotonic) means that new general or specialised concepts should be included in the ontology in a way that does not require the revision of existing definitions [Gómez-Pérez 1996, 2004]. Applying **minimal ontological commitment principle**, which means making as few claims as possible about the world being modelled, gives users of the ontology freedom to specialize and instantiate the ontology as required.

Evaluation of expandability criteria requires inspection of the ontology code. Easily expandable ontologies are also well understandable by users. Definitions of subclasses of the same class are advisable to be defined according to the same patterns to ease the inclusion of new definitions of classes.

For **validation of expandability** of the proposed EU DPP core ontology some examples of possible candidates for subclasses or sub-properties can be drawn from additional information provided by relevant sectors of interest like the Batteries, Construction, and Textile.

7.3 VALIDATION OF THE PROPOSED EU DPP CORE ONTOLOGY

Validation of the proposed EU DPP core ontology is basically performed during development of sector-specific ontologies in WP3 of the CIRPASS-2 project.

The possibility to correctly map the proposed EU DPP core ontology in sector-specific use cases will be validated using specific examples of DPPs. Because the mandatory information models are already well-defined for the Construction and Batteries sectors, these may be used for the general validation of the possibility to map the proposed EU DPP core ontology. However, because of the limited maturity of the mandatory information model for the Textile sector, only the mapping to the LCA part of the proposed EU DPP core ontology may be defined. For these tasks, the following documents will be used.

7.3.1 CONSTRUCTION PRODUCTS

The construction sector has extensive and well-defined data standards. These are extensively described in CIRPASS-2 Deliverable 3.2 “Upper-Level Regulatory Sectoral Ontologies & Vocabularies”¹⁷.

7.3.2 BATTERIES

A data model corresponding to the mandatory information requirements for batteries has been published by the Battery Pass project in [github.com](https://github.com/batterypass/BatteryPassDataModel/tree/main)¹⁸ and its data model is based on a Data Attribute Longlist¹⁹. This data model is currently the basis of ongoing work in DIN DKE SPEC 99100 for the battery passport vocabulary, which may become an ISO standard. The first version of the DIN DKE SPEC 99100 was published in February 2025²⁰.

7.3.3 TEXTILE PRODUCTS

Relevant information requirements for garments are defined in the JRC preparatory study document “Task 4 of the preparatory study on textile products - November 2024”²¹, Annex (page 68). On page 81, section: 2.1.2.15 Environmental impacts, a list of 7 environmental impact categories that are relevant are provided as follows:

1. Water consumption.
2. Energy consumption.
3. The use of chemical substances and mixtures. Some of them are pesticides, solvents, surfactants, dyes and pigments, water and stain repellents, flame retardants, biocides, etc.
4. Textile waste, including post-industrial, pre-consumer and post-consumer waste.
5. Consumption of greenhouse gases.
6. Emissions of microplastics during the fragmentation of synthetic textiles.
7. Chemical Oxygen Demand (COD), Oxides of Nitrogen (NOx), and Oxides of Sulphur (SOx) pollution.

8 DEFINITIONS OF KEY NOTIONS AND CORE TERMS

In MOMo, the identified key notions may become an ontology module, or some key notions together may form a module. In this section, identified key notions are listed as headings of subsections. Each subsection represents an expected module that can be drawn from the corresponding key notion. Currently, the document represents the following key notions (potential modules):

- Product
 - DPP
 - Life Cycle Events
-

¹⁷ <https://zenodo.org/records/14900883>

¹⁸ <https://github.com/batterypass/BatteryPassDataModel/tree/main>

¹⁹ https://thebatteryapp.eu/assets/images/content-guidance/pdf/2023_Battery_Passport_Data_Attributes.xlsx

²⁰ https://thebatteryapp.eu/assets/images/content-guidance/pdf/DIN_DKE_SPEC_99100.PDF

²¹ <https://susproc.jrc.ec.europa.eu/product-bureau/product-groups/467/documents>

- Environmental Impact
- Product compliance
- LCA
- Actors
- Units

Unlike MOMo, we simultaneously identify an initial set of core terms, which are related to the corresponding key notions.

The terms used in the proposed EU DPP core ontology are originating and consumed as defined (i.e., they are taken from the listed documents) in the following acts and documents in the given order (i.e., the increasing order determines from which document the term shall be considered the first (starting from 1)):

1. ESPR²²
2. Standardisation request 5423²³
3. ISO/IEC 17000:2020²⁴
4. ISO/IEC Guide:2007²⁵
5. CIRPASS-2 User Stories 3.0²⁶
6. CIRPASS Glossary²⁷
7. CIRPASS Information Requirements²⁸
8. Terms that are defined by CIRPASS-2 consortium.

In case the term is not defined in any of the documents listed above, or the given definition does not serve the merit of DPP, a new term is defined for the DPP core ontology. In case a term is defined in many documents, the order of the documents is considered first, and then the context of the proposed EU DPP core ontology. Application notes: For example, if a *Term* exists in ESPR and any of the following documents in the list above, the definition provided in the ESPR is considered paramount.

8.1 PRODUCT RELATED TERMS

Product

means any physical goods that are placed on the market or put into service. ESPR Art 2 (1))

Component

²² <http://data.europa.eu/eli/reg/2024/1781/oj>

²³ [https://ec.europa.eu/transparency/documents-register/detail?ref=C\(2024\)5423&lang=en](https://ec.europa.eu/transparency/documents-register/detail?ref=C(2024)5423&lang=en)

²⁴ <https://www.iso.org/standard/73029.html>

²⁵ <https://www.iso.org/standard/45324.html>

²⁶ <https://doi.org/10.5281/zenodo.13902463>

²⁷ This is an unpublished file of terms and definitions collected by the CIRPASS project.

²⁸ https://cirpassproject.eu/wp-content/uploads/2024/06/D2.1_Mapping-of-legal-and-voluntary-requirements-and-Screening-of-emerging-DPP-related-pilots-17-07-2023_new.pdf

means a product intended to be incorporated into another product. (ESPR Art 2 (2))

Intermediate product

means a product that requires further manufacturing or transformation such as mixing, coating or assembling to make it suitable for end-users. (ESPR Art 2 (3))

Energy-related product

means any product that has an impact on energy consumption during use. (ESPR Art 2 (4))

Product group

means a set of products that serve similar purposes and are similar in terms of use, or have similar functional properties, and are similar in terms of consumer perception (ESPR Art 2 (4)). The product group or groups covered are given by delegated acts. (ESPR Art 8 (a))

Consumer product

means any product, excluding components and intermediate products, primarily intended for consumers. (ESPR Art 2 (36))

Product presenting a risk

means a product that, by not complying with an Ecodesign requirement set in or pursuant to this Regulation other than those listed in Article 71(1), could adversely affect the environment or other public interests protected by that requirement. (ESPR Art 2 (57))

Product presenting a serious risk

means a product presenting a risk for which, based on an assessment, the degree of the relevant non-compliance or the associated harm is considered to require rapid intervention by the market surveillance authorities, including cases where the effects of the non-compliance are not immediate. (ESPR Art 2 (58))

Unique product identifier

means a unique string of characters for the identification of a product that also enables a web link to the digital product passport. (ESPR Art 2 (30))

The unique product identifier shall always allow the possibility to include the three different granularity levels, i.e., model, batch, or item. (Standardisation request 5423 Annex II, Part B 1.4).

Technical specification

means a document that prescribes technical requirements to be fulfilled by a product, process or service. (ESPR Art 2 (49))

Batch

means a subset of a specific model composed of all products produced in a specific manufacturing plant at a specific moment in time. (Standardisation request 5423 Annex II, Part B 1.1 (3))

Model

means a version of a product of which all units share the same technical characteristics and the same model identifier. (Standardisation request 5423 Annex II, Part B 1.1 (2))

Item

means a single unit of a model. (Standardisation request 5423 Annex II, Part B 1.1 (4))

Digital instructions

means instructions in digital format concerning the product ('digital instructions') in a language that can be easily understood. (ESPR Art 27(7))

Digital instructions are ensured by manufacturers. These should also be ensured by importers. (ESPR Art 29(4))

Instructions for the safe use of the product. (ESPR Art 7 (5), (d))

Repair and maintenance instructions. (ESPR Annex I (b))

Performance requirement

means a quantitative or non-quantitative requirement for or in relation to a product to achieve a certain performance level in relation to a product parameter referred to in Annex I. (ESPR Art 2 (8))

Class of performance

means a range of performance levels in relation to one or more product parameters referred to in Annex I, which is established based on a common methodology for the product or product group, ordered in such a way as to allow for product differentiation. (ESPR Art 2 (15))

Product parameters

are listed in the ESPR Annex I as follows:

- (a) durability and reliability;
- (b) ease of repair and maintenance;
- (c) ease of upgrading, reuse, remanufacturing and refurbishment;
- (d) design for recycling, ease and quality of recycling;
- (e) avoidance of technical solutions detrimental to reuse, upgrading, repair, maintenance, refurbishment, remanufacturing and recycling of products and components;
- (f) use of substances, and in particular the use of substances of concern;
- (g) use or consumption of energy, water and other resources;
- (h) use or content of recycled materials and recovery of materials, including critical raw materials;
- (i) use or content of sustainable renewable materials;
- (j) weight and volume of the product and its packaging, and the product-to-packaging ratio;
- (k) incorporation of used components;
- (l) quantity, characteristics and availability of consumables needed for proper use and maintenance;
- (m) the environmental footprint of the product, expressed as a quantification;
- (n) the carbon footprint of the product;
- (o) the material footprint of the product;
- (p) microplastic and nano-plastic release as expressed through the release during relevant product life cycle stages, including manufacturing, transport, use and end-of-life stages;

- (q) emissions to air, water or soil released in one or more lifecycle stages of the product as expressed through quantities and nature of emissions, including noise;
- (r) amounts of waste generated, including plastic waste and packaging waste and their ease of reuse, and amounts of hazardous waste generated;
- (s) functional performance and conditions for use;
- (t) lightweight design;

Durability

means the ability of a product to maintain over time its function and performance under specified conditions of use, maintenance and repair. (ESPR Art 2 (22))

Reliability

means the probability that a product functions as required under given conditions for a given duration without an occurrence which results in a primary or secondary function of the product no longer being performed. (ESPR Art 2 (16))

Distance selling

means the offer for sale, hire or hire purchase of products, online or through other means of distance sales, whereby the potential customer cannot physically access the product. (ESPR Art 2 (56))

8.2 PRODUCT CONFORMITY RELATED TERMS

Quality information

means information on the quality of the product (model) that has been attested by assessment. Information on assessment provides all kinds of statements on conformity to standards, regulations, (voluntary) product labels, implementing acts, etc. (attestation data). These standards refer to domains of conformity in eco-design, governmental requirements, social requirements and all other aspects of product quality that could be relevant. It must be visible if a product has quality information provided or not.

Conformity attestation (compliance documentation)

means a formal document or declaration issued by a party undertaking an assessment of a product, system, or process, stating that compliance with specific standards, regulations, or requirements has been demonstrated. These attestations are compliance documentation and information required under this Regulation or other Union law applicable to the product, such as the declaration of conformity, technical documentation or conformity certificates (ESPR Annex III (e)). Core data of conformity attestation are the type of attestation, its status, the period of time of validity, involved parties and their roles, links to attestation files and data on the conformity assessment process.

Attestation type

means a classification of the attestation such as certification, declaration, testing, inspection, verification, validation, calibration, compliance report, etc.

Attestation status

means information on the validity of the attestation such as pending, valid, expired, revoked, etc.

Attestation validity

means a specific period of time in which the attestation is valid.

Attestation start date

means the date when the validity of the attestations begins.

Attestation end date

means the date when the validity of the attestations ends.

Attestation party

means the parties that are involved in attestation and related assessment activities such as issuer, recipient, third-party accreditation of assessment process, etc.

Attestation party role

means the classification of role a party is taking in respect of the attestation. Roles are for example authority, regulatory approved authority, accreditation authority, producer/manufacturer, economic operator, accreditation issuing authority, conformity assessment body, etc.

Conformity assessment body

means a body that performs conformity assessment activities including calibration, testing, certification and inspection, excluding accreditation. (ESPR Art 2 (52))

Conformity assessment

means the process demonstrating whether the eco-design requirements set out in the relevant delegated acts adopted pursuant to Article 4 have been fulfilled. (ESPR Art 2 (51), ISO/IEC 17000:2020 sec. 4.1). Conformity assessment is made concerning requirements and rules set by relevant regulations, standards, and other relevant technical specifications (e.g., for product labels).

Regulation

means a rule or directive made and maintained by an authority. ESPR itself is a regulation issued by the Commission of the European Union, it is part of the New Legislative Framework, and several delegated acts detail regulatory requirements for categories of different products.

Standard

means and technical specification agreed by experts within a normative body. Standards exist on the national, regional (European) and international level.

Label

means information on specific (voluntary) labels applicable to the product. That shall include whether an EU Ecolabel has been awarded to the product in line with Regulation (EC) No 66/2010. (ESPR Annex III).

EU Ecolabel

means the label provided by the European union that promotes goods and services with a guaranteed reduced environmental impact throughout their entire life cycle information on specific (voluntary) labels applicable to the product. (EC No 66/2010)

EU Declaration of conformity

means European declaration of conformity (ESPR Annex V). It is an attestation document and commonly the outcome of a self-declaration by manufacturer for example.

CE marking

means a marking by which the manufacturer indicates that the relevant product is in conformity with the applicable requirements set out in Union harmonisation legislation providing for its affixing. (ESPR Art 2 (50))

Conformity marking

means markings of products other than the CE marking following rules pursuant to ESPR Article 4(6), point (d), for its affixing.

8.3 ENVIRONMENTAL IMPACT RELATED TERMS

Environmental impact

means any change to the environment, whether adverse or beneficial, wholly or partially resulting from a product during its life cycle. (ESPR Art 2 (14))

Environmental footprint

means a quantification of the environmental impacts resulting from a product throughout its life cycle, whether in relation to a single environmental impact category or an aggregated set of impact categories based on the Product Environmental Footprint method established by Recommendation (EU) 2021/2279 or other scientific methods developed by international organisations, widely tested in collaboration with different industry sectors and adopted or implemented by the Commission in other Union law. (ESPR Art 2 (24))

Carbon footprint

means the sum of greenhouse gas emissions and greenhouse gas removals in a product system, expressed as CO₂ equivalents and based on a life cycle assessment using the single impact category of climate change. (ESPR Art 2 (25))

Material footprint

refers to the total amount of raw materials extracted to meet final consumption demands. (ESPR Art 2 (26))

Substance

means a chemical element and its compounds in the natural state or obtained by any manufacturing process, including any additive necessary to preserve its stability and any impurity deriving from the process used, but excluding any solvent which may be separated without affecting the stability of the substance or changing its composition (ESPR Art 2(27)).

Substance of concern

means a substance that meets the criteria laid down in Article 57 of Regulation (EC) No 1907/2006 and is identified in accordance with Article 59 (1) of that Regulation or meets other criteria given in ESPR Art 2 (27).

Location of substance of concern

within the product (ESPR Art 7 (5b)).

Concentration of the substance of concern

the concentration of the substances of concern, at the level of the product, its relevant components, or spare parts (ESPR Art 7(5)).

Maximum concentration of the substance of concern

the maximum concentration of the substances of concern, at the level of the product, its relevant components, or spare parts (ESPR Art 7(5)).

Concentration range of the substance of concern

at the level of product (ESPR Art 7 (5c)) and at the level of product' relevant components (ESPR Art 7 (5c)).

8.4 PRODUCT LIFE CYCLE EVENTS AND RELATED TERMS

Putting into service

means the first use, for its intended purpose, in the Union, of a product. (Regulation (EU) 2019/1020, Art 3 (22))

Life cycle

means the consecutive and interlinked stages of a product's life, consisting of raw material acquisition or generation from natural resources, pre-processing, manufacturing, storage, distribution, installation, use, maintenance, repair, upgrading, refurbishment and reuse, and end-of-life. (ESPR Art 2 (12))

Destruction

means the intentional damaging or discarding of a product as waste with the exception of discarding for the sole purpose of delivering the discarded product for preparing for reuse, including refurbishment or remanufacturing operations. (ESPR Art 2 (34))

End-of-life

means the life cycle stage that begins when a product is discarded and ends when the waste material of the product is returned to nature or enters another product's life cycle. (ESPR Art 2 (13))

Maintenance

means one or more actions carried out to keep a product in a condition where it is able to fulfil its intended purpose. (ESPR Art 2 (19))

means an action carried out to keep a product in a condition where it is able to function as required (Standardisation request 5423 Annex II, Part A (8))

Making available on the market

means any supply of a product for distribution, consumption or use on the Union market in the course of a commercial activity, whether in return for payment or free of charge. (ESPR Art 2 (39))

Placing on the market

means the first making available of a product on the Union market. (ESPR Art 2 (40))

Premature obsolescence

means a product design feature or subsequent action or omission resulting in the product becoming non-functional or performing less well without such changes of functionality or performance being the result of normal wear and tear. (ESPR Art 2 (21))

Preparing for reuse

means checking, cleaning or repairing recovery operations, by which products or components of products that have become waste are prepared so that they can be re-used without any other pre-processing. (Directive 2008/98/EC, Art 3 (16))

Recall

means any measure aimed at achieving the return of a product that has already been made available to the end user. (Directive 2008/98/EC, Art 3 (15))

Recovery

means any operation the principal result of which is waste serving a useful purpose by replacing other materials which would otherwise have been used to fulfil a particular function, or waste being prepared to fulfil that function, in the plant or in the wider economy. Annex II sets out a non-exhaustive list of recovery operations. (Directive 2008/98/EC, Art 3 (17))

Recycling

means any recovery operation by which waste materials are reprocessed into products, materials or substances whether for the original or other purposes. It includes the reprocessing of organic material but does not include energy recovery and the reprocessing into materials that are to be used as fuels or for backfilling operations. (ESPR Art 2 (18))

Refurbishment

means actions carried out to prepare, clean, test, service and, where necessary, repair a product or a discarded product in order to restore its performance or functionality within the intended use and range of performance originally conceived at the design stage at the time of the placing of the product on the market. (Standardisation request 5423 Annex II, Part A (7))

means preparing or modifying an object that is waste or a product to restore its performance or functionality within the intended use, range of performance and maintenance originally conceived at the design stage, or to meet applicable technical standards or regulatory requirements, with the result of making a fully functional product. (Regulation (EU) 2019/1020, Art 3 (25))

Remanufacturing

means actions through which a new product is produced from objects that are waste, products or components and through which at least one change is made that substantially affects the safety, performance, purpose or type of the product. (ESPR Art 2 (20))

Repair

means one or more actions carried out to return a defective product or waste to a condition where it fulfils its intended purpose. (Standardisation request 5423 Annex II, Part A (9))

Reuse

means any operation by which products or components that are not waste are used again for the same purpose for which they were conceived. (Directive 2008/98/EC, Art 3 (13))

Upgrading

means actions carried out to enhance the functionality, performance, capacity, safety or aesthetics of a product. (Standardisation request 5423 Annex II, Part A (6) and ESPR Art 2 (17))

Value chain

means all activities and processes that are part of the life cycle of a product, as well as its possible remanufacturing. (ESPR Art 2 (11))

Waste

means any substance or object which the holder discards or intends or is required to discard. (Regulation (EU) 2019/1020, Art 3 (23))

8.5 GENERIC LIFE CYCLE IMPACT ASSESSMENT (LCIA) RELATED TERMS

The following terms originate from the Life Cycle Impact Assessment (LCIA) Module of an LCA ontology²⁹.

Methodology

Related LCIA methods can be grouped under certain LCIA methodologies. Thus, a methodology contains multiple methods. A methodology (identified by an LCA expert for carrying out a specific LCA study), assesses some topic which is included in the scope of the LCA study. (ILCD Handbook – General guidance)

Method

LCIA methods allow to transform raw emissions and resource data into understandable impact potential scores. Examples of methods include EN15804+A2, EF v3.1, Impact World.

Characterization Model (CM)

A characterization model provides a mathematical framework, as a set of equations, to quantify an impact category based on the chosen method. Example: IPCC (2021) AR6. A characterization model is used (imposed by) a method.

Characterization Factor (CF)

Impact factors (aka characterization factors): Factors that quantify the relative impact of different substances within the same category. These factors are used to convert the amount of emissions into impact potential scores for each category.

Impact Category (IC)

An impact category represents a specific environmental issue that the LCA aims to assess, such as global warming or ozone depletion. Each category aggregates the effects of different emissions that contribute to the same environmental problem.

Impact Category Indicator (ICI)

An impact category refers to a specific metric/measurable quantity that is used to quantify an impact category, such as (as “midpoint impact potential” examples): Global Warming Potential (GWP) and Ozone Depletion Potential (ODP).

Impact (potential) result

An impact result corresponds to the computed value of each flow (or sometimes activity as a group of its input and output flows) by the ICI. Thus, it is an entity that links an impact category indicator (e.g. GWP-100) to a product reference flow (e.g. a unit of a specific flat chip resistor) of a single component of a more complex product, to its exact value (e.g. 1.16E-03 in the unit kg CO₂-equivalents). This value is also called “carbon footprint” in this example.

Overall impact (potential) result

²⁹ <https://github.com/DanashFatima/LCA>

It corresponds to the sum of all impact result values of all flows/activities in the studied LCA of product X, regarding an impact category Y.

8.6 DPP RELATED TERMS

Digital product passport

means a set of data specific to a product that includes the information specified in the applicable delegated act adopted pursuant to Article 4 and that is accessible via electronic means through a data carrier in accordance with Chapter III. (ESPR Art 2 (28))

Where a new digital product passport is created for a product that already has a digital product passport, the new digital product passport shall be linked to the original digital product passport or passports. (ESPR Art 11 (d))

The product passport should also allow the inclusion of information concerning integrated product components, where relevant. (Standardisation Request 5423 3))

The product passport should be defined at the level of item, batch or product model, depending on specific needs and complexity of the value chain, the size, nature or impacts of the products considered. (Standardisation Request 5423 7))

The product passport should be linked to a unique product identifier. Standardisation Request 5423 (8))

DPP should include user manuals, instructions, warnings or safety information, as required by other Union law applicable to the product. (ESPR Annex 3 (f))

Data carrier

means a linear barcode symbol, a two-dimensional symbol or other automatic identification data capture medium that can be read by a device. (ESPR Art 2 (29))

DPP-data

means the information included in a product passport and accessible to different users based on their own respective access rights. (Standardisation Request 5423, Annex II (16))

EU Registry

means the digital product passport registry to be set up by the European Commission to store in a secure manner at least the unique identifiers linked to products placed on the market or put into service in the EU (ESPR Art 13 (1)).

Unique registration identifier

means a unique registration identifier associated with the unique identifiers uploaded into the EU registry for a specific product in accordance with ESPR Art 13 (4) by the economic operator placing the product on the market or putting it into service.

Public authority

Means legal person, who stores the necessary DPP information in the registry and issues a **unique registration identifier**. (CIRASS-2 User Stories 3.0, Glossary)

8.7 ACTORS RELATED TERMS

Consumer

means any natural person who, in relation to contracts covered by this Directive, is acting for purposes which are outside that person's trade, business, craft or profession. (Directive (EU) 2019/771, Art 2 (2))

Customer

means a natural or legal person that purchases, hires or receives a product for their own use whether or not acting for purposes which are outside their trade, business, craft or profession. (ESPR Art 2 (35))

Customs authorities

means customs authorities as defined in point 1 of Article 5 of Regulation (EU) No 952/2013 (Regulation (EU) 2019/1020, Art 3 (24))

Manufacturer

means any natural or legal person that manufactures a product or that has a product designed or manufactured and markets that product under their name or trademark. (ESPR Art 2 (42))

Authorised representative

means any natural or legal person established in the Union that has received a written mandate from the manufacturer to act on the manufacturer's behalf in relation to specified tasks with regard to the manufacturer's obligations under this Regulation. (ESPR Art 2 (43))

Importer

means any natural or legal person established in the Union that places a product from a third country on the Union market. (ESPR Art 2 (44))

Distributor

means any natural or legal person in the supply chain, other than the manufacturer or the importer, that makes a product available on the market. (ESPR Art 2 (45))

Economic operator

means the manufacturer, the authorised representative, the importer, the distributor, the dealer and the fulfilment service provider. (ESPR Art 2 (46))

Fulfilment service provider

means any natural or legal person offering, in the course of commercial activity, at least two of the following services: warehousing, packaging, addressing and dispatching, without having ownership of the products involved, excluding postal services as defined in point 1 of Article 2 of Directive 97/67/EC of the European Parliament and of the Council (31), parcel delivery services as defined in point 2 of Article 2 of Regulation (EU) 2018/644 of the European Parliament and of the Council (32), and any other postal services or freight transport services. Regulation (EU) 2019/1020.

Independent operator

means a natural or legal person that is independent of the manufacturer and is directly or indirectly involved in the refurbishment, repair, maintenance or repurposing of a product, and includes waste management operators, refurbishers, repairers, manufacturers or distributors of repair equipment, tools or spare parts, as well as publishers of technical information, operators offering inspection and

testing services and operators offering training for installers, manufacturers and repairers of equipment. (ESPR Art 2 (47))

Professional repairer

means a natural or legal person that provides professional repair or maintenance services for a product, irrespective of whether that person acts within the manufacturer's distribution system or independently. (ESPR Art 2 (48))

Dealer

means a distributor or any other natural or legal person that offers products for sale, hire or hire purchase, or that displays products, to end users in the course of a commercial activity, including through distance selling; and includes any natural or legal person that puts a product into service in the course of a commercial activity. (ESPR Art 2 (55))

Unique operator identifier

means a unique string of characters for the identification of an actor involved in a product's value chain. (ESPR Art 2 (31))

Digital product passport service provider

means a natural or legal person that is an independent third-party authorised by the economic operator which places the product on the market or puts it into service and that processes the digital product passport data for that product for the purpose of making such data available to economic operators and other relevant actors with a right to access those data under this Regulation or other Union law. (ESPR Art 2 (32))

Unique facility identifier

means a unique string of characters for the identification of locations or buildings involved in a product's value chain or used by actors involved in a product's value chain. (ESPR Art 2 (33))

Data provider

is the economic operator responsible for providing the information to the economic operator responsible for issuing the DPP. (CIRPASS D2.1 (3.2.1))

Responsible Economic Operator (REO)

refers to a legal person that is responsible for creating a DPP, and all legal obligations therewith. (CIRPASS-2, User stories 3.0, page 13)

Only one economic operator can be the REO under the ESPR for a product at a given time. (CIRPASS-2, User Stories 3.0, page 63)

See also, APPENDIX B: A NOTE ON 'RESPONSIBLE' ECONOMIC OPERATORS CIRPASS-2, User stories 3.0, page 83). "Thus, it will be an economic operator, or their authorised digital product passport service provider that is responsible for the creation of a DPP." and "The inference that we make from the above cited articles for the purposes of this document, is that the Responsible Economic Operator is likely the manufacturer, the importer, or their authorized representative."

Issuing agency

are legal persons that provide unique identifiers and data carriers Art 12 (4a). Under specific conditions, economic operators may perform the functions associated with an Issuing Agency. (ESPR Art 12 (4b))

Credential agency

means a legal person that provides (professional) credentials to parties, which may be used to make and verify a variety of claims in the DPP. Definition is derived from EUPR Art 11 last paragraph “The Commission may adopt implementing acts setting out procedures to issue and verify the digital credentials of economic operators and other relevant actors that have access rights to data included in the digital product passport”. The credentials agency can provide all credentials except for the unique identifiers and data carriers, which are provided by an issuing agency. (CIRPASS-2 User Stories 3.0, Glossary, p 80)

Actor

means a legal or natural person. One actor can take on multiple roles. CIRPASS-2 User Stories 3.0, Glossary, p 80)

Value chain actors

are economic operators, namely manufacturers, authorised representatives, importers, distributors, dealers and fulfilment service providers, and other value chain actors, such as customers, professional repairers, independent operators, refurbishers, remanufacturers, recyclers, market surveillance and customs authorities, civil society organisations, researchers, trade unions, and the Commission, or any organisation acting on their behalf. (EUPR recital 32)

Role

means a set of tasks typically performed by one actor. (e.g. Responsible Economic Operator, Independent operator). (CIRPASS-2 User Stories 3.0)

Product passport access rights

mean that rights to introduce, modify or update data in the digital product passport shall be restricted based on the access rights specified in delegated acts adopted. (EUPR Art 11.1 (f) p41)

Access rights management shall be attribute-based. (Standardisation request 5423 ANNEX II, Part B 3.2) p9)

Economic operators shall be responsible for managing the corresponding product passport access rights. (Standardisation request 5423 ANNEX II Part B 3.2) p9)

Access rights for each information included in the product passport shall be product group specific. (Standardisation request 5423 ANNEX II, Part B (3.2) p9)

Public data included in the product passport shall not require any access right management. (Standardisation request 5423 ANNEX II, Part B (3.2) p9)

Accreditation

In scope of EUPR, accreditation is convened with attestation by a national accreditation body that a conformity assessment body meets the requirements set by harmonised standards and, where applicable, any additional requirements including those set out in relevant sectoral schemes, to carry out a specific conformity assessment activity. (Regulation (EC) No 765/2008, Art 2 (10))

Corrective action

means any action taken by an economic operator to bring any non-compliance to an end where required by a market surveillance authority or on the economic operator's own initiative. (Regulation (EU) 2019/1020, Art 3 (16))

8.8 MEASUREMENT UNIT AND QUANTITY RELATED TERMS

Measurand

A measurand (respectively a property) of the product (model) is the quantity intended to be measured (to be expressed). The measurand is expressed by a quantity value. Product parameters such as weight and packaging size of a product are measurands. (VIM 2.3: The quantity intended to be measured. The specification of a measurand requires knowledge of the kind of quantity, description of the state of the phenomenon, body, or substance carrying the quantity, including any relevant component, and the chemical entities involved.)

Quantity value

Expression of the magnitude of a quantity/property by a number (numerical value) and a reference in the form of a unit, a reference material, or a reference procedure. (VIM 1.19: number and reference together expressing magnitude of a quantity)

Quantity kind

Statement of kind of quantity such as length, mass, time, temperature, concentration of substance, (VIM 1.2: aspect common to mutually comparable quantities)

Value

The numerical value of the quantity. (VIM: 1.20 number in the expression of a quantity value, other than any number serving as the reference)

Measurement unit

A measurement unit such as kilogram, metre, etc., provided as a reference for the magnitude of the quantity in respect to its value. (VIM 1.9: real scalar quantity, defined and adopted by convention, with which any other quantity of the same kind can be compared to express the ratio of the two quantities as a number)

Standard measurement uncertainty

The standard measurement uncertainty provides information on the accuracy of a quantity value. It is expressed by a positive numerical value of a standard deviation and an information on the statistical distribution of the quantity values from a series of measurements. (VIM 2.30: Measurement uncertainty expressed as a standard deviation.)

Expanded measurement uncertainty

The expanded measurement uncertainty provides information on the accuracy of a quantity value. It is a product of a combined standard measurement uncertainty, and a coverage factor larger than one spanning a symmetric interval with a defined coverage probability of quantity values from a series of measurements. (VIM 2.35: product of a combined standard measurement uncertainty and a factor larger than the number one)

9 ONTOLOGY REQUIREMENTS GAPS

The proposed EU DPP core ontology requirements (basically DPP information requirements) are discussed in the ESPR, Standardisation request 5423 and several deliverables of the CIRPASS project. However, the following gaps for ontology development exist:

1. Currently, the terms “Natural person” and “Legal person” are not defined. This influences concepts like Dealer, Professional repairer and Independent operator as they are defined in the ESPR having

the following subclass relationships: Dealer <is-a> Natural person, Dealer <is-a> Legal person; Professional repairer <is-a> Natural person, Professional repairer <is-a> Legal person; Independent operator <is-a> Natural person, Independent operator <is-a> Legal person. This in turn influences the overall hierarchy of the Economic operator concept. The issue will be solved in the design phase of the proposed EU DPP core ontology. Probably Actors and Roles concepts are introduced to the ontology as a solution.

2. Lack of consistency and clarity in the terminology concerning the notion of data model related to the proposed EU DPP core ontology concept Product and its properties in the CIRPASS-2 User Stories 3.0 document. In particular, the term “product data model” used in CIRPASS-2 User Stories 3.0 (User Story 1) is not defined. However, assumptions are given as follows “Public Authorities facilitate a standard machine-readable data model for products that can be requested from the relevant authorities’ server. This data model may differ depending on the relevant delegated act.” (CIRPASS-2 User Stories 3.0, p 27). Second, the term “DPP data model” is used in CIRPASS-2 User Stories 3.0 document User story 1, 2. as follows: “Public Authorities ensure availability of a service (e.g., by providing the service themselves or ensuring that others provide it) via which a standard data model for products regulated via a delegated act is made available. Making available here means at least that it can be requested from the relevant authorities’ server in a standard machine-readable format (e.g., SHACL, CSV, JSON, XML, etc.)”. Third, in CIRPASS-2 User Stories 3.0, p. 80 the term “data model” is defined as follows: “Data model means a “structured representation of data elements and relationships used to facilitate semantic interoperability within and across domains” as defined by DSSC³⁰”. In DSSC, “Data models: common languages to facilitate semantic interoperability in a data space, including ontologies, data models, schema specifications, mappings and API specifications that can be used to annotate and describe data sets and data services. Such vocabularies are often domain specific.” Consequently, ontologies are a kind of data model in DSSC, which is an incorrect statement. In addition, CIRPASS-2 User Stories 3.0 (p. 90) uses the term “common cross-sectoral data model”, which is not defined.

10 RECOMMENDATIONS

To make definitions consistent and clear, as well as applicable, we clarify some of the terms related to data model and ontology related to the proposed EU DPP core ontology development.

A data model is a structured representation (often informal or semi-formal) of data elements, their attributes, and the relationships between them. It enables efficient storage, retrieval, and manipulation of information [Chen 1976, Silberschatz et al., 2020].

An ontology is a formal representation that defines concepts, their meanings, attributes, and interrelations within a domain of knowledge. It captures the semantics of a domain using logical formalisms, allowing for reasoning, inference, and the alignment of data across systems [Gruber 1993].

Note: The statement “Ontology is a data model” used in DSSC implies that ontologies are a subset of data models, which can lead to confusion. While both are used to describe data and its organization, their purposes and scope differ significantly.

Ontologies are not just another type of data model – they add a layer of semantics and reasoning capabilities that traditional data models lack of. Additionally, the definition “structured representation of data elements and relationships used to facilitate semantic interoperability” (CIRPASS-2 User Stories 3.0, p 8) is more aligned with a purpose of an ontology than a data model.

³⁰ <https://dssc.eu/space/BVE/357075098/Data+Models>

11 POTENTIAL FUTURE REQUIREMENTS FOR THE PROPOSED EU DPP CORE ONTOLOGY

Potential future requirements for the development of the proposed EU DPP core ontology have been identified as follows:

1. The ESPR and forthcoming delegated acts may require any physical good, including components and intermediate products, to be placed on the EU market or put into service, to have the EU DPP. As this requirement covers products not manufactured only in the EU but also exported into the EU, it has an impact on global trade. Thus, all countries who export goods to the EU should take care that exported goods will be equipped with DPPs compliant to the EU DPP.

To support governments and industry with practical measures to counter greenwashing by implementing supply chain traceability and transparency the United Nations Economic Commission for Europe (UNECE) proposed the UN Transparency Protocol (UNTP)³¹.

The UNTP defines a standard and extensible vocabulary of Environmental, Social and Governance (ESG) criteria and provides a mechanism for any standards authority, or national regulator, or industry association to map their specific terminology to the UNTP vocabulary. **UNTP uses JSON-LD for creating web vocabularies** and meaningful transparency graphs. The UNTP is under development. The test version of UNTP Core Vocabulary is represented in JSON-LD.

The UNTP DPP does not conflict with the concept of the EU DPP. It provides the data about supply-chain activities needed for the issuing of the EU DPPs. Detailed comparison of the EU DPP and the UNTP DPP is found in the CIRPASS document “DPP Prototypes”³².

United Nations Centre for Trade Facilitation and Electronic Business (UN/CEFACT) has developed a semantic interoperability framework for verifiable credentials based on **JSON-LD vocabularies**. The UN/CEFACT Web Vocabularies are available at <https://vocabulary.uncefact.org/>.

From the perspective of semantic interoperability, the synergies with UNTP may be explored in future.

2. In future developments of the proposed EU DPP core ontology, it is valuable to attach versioning and data quality metrics to data that is used for instantiation of the ontologies related to the proposed EU DPP core ontology. Information about data quality and versioning can be added by reusing the Provenance Ontology PROV-O³³ and the Data Quality Vocabulary (DQV)³⁴.

³¹ <https://uncefact.github.io/spec-untp/docs/about/>

³² <https://cirpassproject.eu/project-results/>

³³ <http://www.w3.org/TR/prov-o/>

³⁴ <https://www.w3.org/TR/vocab-dqv/>

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